ESE 4011 One-Shot Notes

Materials for Energy Engineering Application · KUET · past papers 2022 · answers from standard textbooks (no course slides available)

One-shot notes: ESE 4011 Materials for Energy Engineering Application

Solar cell losses

Optical: reflection (bare Si 30-40 %, fix with ARC λ/4n + texture), shading by fingers (5-10 %), transmission (light trapping, back reflector).

Spectral: non-absorption ($E < E_g$) and thermalisation ($E - E_g$ as heat). **Recombination**: radiative, Auger, SRH, surface (passivate). **Resistive**: series and shunt (fill factor).

c-Si and devices

Mono (one crystal, Cz/FZ, 20-26 %), multi (grains, cast, 16-22 %), amorphous a-Si:H (no order, PECVD, E_g 1.7 eV, 6-10 %).

Diffusion device = p-n, pure material, carriers diffuse (c-Si). **Drift device** = p-i-n, field in i-layer sweeps carriers (a-Si).

PERL (UNSW, 25 %)

Inverted pyramids, double ARC on thin oxide, light n^+ emitter + heavy n^{++} under fingers, rear oxide passivation + reflector, local p^+ point contacts, FZ wafer. Led to PERC, PERT, TOPCon, IBC.

Thin films

DSSC: FTO / dye on mesoporous TiO_2 / I^-/I_3^- electrolyte / Pt counter. Dye excites → injects e^- into TiO_2 → load → Pt reduces I_3^- → I^- regenerates dye.

Chalcopyrite I-III-VI_2: $CuInSe_2$ 1.0, $CuGaSe_2$ 1.7, $CuInS_2$ 1.5 eV, CIGS 1.1-1.2 eV.

CdTe (superstrate): glass / TCO / buffer / n-CdS (2.4 eV) / p-CdTe (1.45 eV, $CdCl_2$ treated) / back contact.

Beyond one junction

Multijunction: GaInP/GaAs/Ge, gaps decrease downward; > 47 % with concentration. Challenges: current matching, spectrum change, lattice matching, tunnel junctions, cost.

Spectral conversion: down-conversion (1 → 2 photons, cuts thermalisation), down-shifting (UV → visible), up-conversion (2 IR → 1, uses sub-gap light).

Fuel cell basics

Converts fuel chemical energy to electricity continuously; battery stores. Anode: $H_2 \rightarrow 2H^+ + 2e^-$; cathode: $½O_2 + 2H^+ + 2e^- \rightarrow H_2O$. E^0 = 1.23 V, real 0.6-0.8 V.

Stack: series via bipolar plates, gases in parallel via manifolds, end plates + tie rods. **Bipolar plate**: connect, distribute, separate, drain water, cool, support.

PTFE in AFC: hydrophobic, stops flooding, keeps three-phase boundary, binder.

Fuel cell types table

Type	Electrolyte	Ion	T (°C)	Catalyst
PEMFC	Nafion	H^+	60-80	Pt
DMFC	Nafion	H^+	60-120	Pt-Ru / Pt
AFC	KOH	OH^-	60-200	Ni, Ag
PAFC	H_3PO_4 in SiC	H^+	150-220	Pt/C
MCFC	Li/K carbonate in $LiAlO_2$	CO_3^{2-}	650	Ni
SOFC	YSZ ceramic	O^{2-}	600-1000	Ni-YSZ / LSM

DMFC, MCFC, PAFC, SOFC

DMFC layers: plate / GDL / Pt-Ru / Nafion / Pt / GDL / plate. $CH_3OH + H_2O \rightarrow CO_2 + 6H^+ + 6e^-$. Problem: methanol crossover.

MCFC: anode $H_2 + CO_3^{2-} \rightarrow H_2O + CO_2 + 2e^-$; cathode $½O_2 + CO_2 + 2e^- \rightarrow CO_3^{2-}$; CO_2 recycled.

PAFC acid: proton conductor, stable, low volatility, CO_2 tolerant, freezes at 42 °C. Electrodes: Pt/C + PTFE on carbon paper.

SOFC: efficient, fuel flexible, no Pt, all solid, useful heat. Electrolytes: YSZ, ScSZ, GDC/SDC, LSGM, proton perovskites.

Supercapacitors

EDLC: porous carbon + electrolyte; ions form a double layer (d < 1 nm), huge A → C = εA/d large. Electrostatic, 10^6 cycles. $E = ½CV^2$.

Pseudocapacitor: fast surface redox (RuO_2, MnO_2, PANI, PPy); adsorption, redox, intercalation.

Numerical: series $C_{eq} = 120/23 = 5.217$ F, Q = 260.87 C, $V_{15F} = 17.39$ V, $V_{8F} = 32.61$ V; parallel 23 F, 50 V each.

H_2 in carbon

Physisorption (weak, 77 K, ~1 wt % per 500 m^2/g), chemisorption (C-H, needs heat), spillover (Pt/Pd). Increase: area, pores 0.6-0.7 nm, B/N doping, metal decoration, cool. CNT sites: outside, inside, interstitial, grooves; < 1 wt % at room T.

Thin films

Sputtering: vacuum, Ar, plasma, Ar^+ hits target, atoms ejected, deposit. DC (conductors), RF (insulators), magnetron (fast), reactive.

CVD steps: transport, diffuse, adsorb, react, desorb. $SiH_4 \rightarrow Si + 2H_2$. PVD = physical, line-of-sight, low T; CVD = chemical, conformal, higher T, toxic gases.

XRD

$n\lambda = 2d \sin\theta$; $d = a/\sqrt{(h^2+k^2+l^2)}$. Cu Kα 1.5418 Å, Kβ 1.392 Å; **Ni filter** (edge 1.488 Å) or crystal monochromator.

Co (111): d = 2.0461 Å, θ = 22.13°, 2θ = 44.27°.

FCC: $F = f[1 + e^{\pi i(h+k)} + e^{\pi i(h+l)} + e^{\pi i(k+l)}]$: unmixed → 4f, mixed → 0. Present 111, 200, 220, 311, 222. BCC: h+k+l even.

Diffractometer: tube, slits, sample at θ, detector at 2θ, focusing circle.

Electron microscopy

SE: < 50 eV, 1-10 nm deep, topography, Everhart-Thornley. **BSE**: high energy, elastic, Z contrast. SEM: gun, condenser, scan coils, objective, sample, detectors.

TEM: intermediate lens on image plane = image; on back focal plane = diffraction (SAED). Bright field / dark field via objective aperture.

Ceramic prep: saw, 3 mm disc, grind 100 µm, dimple 10-20 µm, Ar ion mill, carbon coat (or FIB).

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
µc-Si:H	hydrogenated microcrystalline silicon
a-Si:H	hydrogenated amorphous silicon
AFC	alkaline fuel cell
ALD	atomic layer deposition
APCVD	atmospheric-pressure CVD
ARC	anti-reflection coating
BSE	backscattered electrons
c-Si	crystalline silicon
CGS	copper gallium selenide
CHP	combined heat and power
CIGS	copper indium gallium (di)selenide, $Cu(In,Ga)Se_2$
CIS	copper indium (di)selenide, $CuInSe_2$
CNT	carbon nanotube
CO	carbon monoxide
CV	cyclic voltammetry
CVD	chemical vapour deposition
Cz	Czochralski (crystal pulling)
CZTS	copper zinc tin sulphide (kesterite)
DC	direct current
DMFC	direct methanol fuel cell
DOE	US Department of Energy
DSSC	dye-sensitised solar cell
EBSD	electron backscatter diffraction
EDLC	electric double-layer capacitor
EDS	energy-dispersive X-ray spectroscopy
EDX	energy-dispersive X-ray (analysis)
Eg	band gap energy
ESR	equivalent series resistance
eV	electron-volt (energy unit, 1.6×10^{-19} J)
FEG	field-emission gun
FIB	focused ion beam
FTO	fluorine-doped tin oxide
FZ	float zone (refining)
GDC	gadolinium-doped ceria
GDL	gas diffusion layer

HIT	heterojunction with intrinsic thin layer
HRTEM	high-resolution TEM
IBC	interdigitated back contact
ICDD	International Centre for Diffraction Data
IR	infrared
ITO	indium tin oxide
IUPAC	International Union of Pure and Applied Chemistry
JCPDS	Joint Committee on Powder Diffraction Standards
LPCVD	low-pressure CVD
LSCF	lanthanum strontium cobalt ferrite
LSGM	strontium- and magnesium-doped lanthanum gallate
LSM	strontium-doped lanthanum manganite
mc-Si	multicrystalline silicon
MCFC	molten carbonate fuel cell
MEA	membrane electrode assembly
MOCVD	metal-organic CVD
MOF	metal-organic framework
MWCNT	multi-walled carbon nanotube
PAFC	phosphoric acid fuel cell
PANI	polyaniline
PCFC	proton-conducting ceramic fuel cell
PECVD	plasma-enhanced chemical vapour deposition
PEDOT	poly(3,4-ethylenedioxythiophene)
PEM	proton exchange (polymer electrolyte) membrane
PEMFC	proton exchange membrane fuel cell
PERC	passivated emitter and rear cell
PERL	passivated emitter, rear locally-diffused (cell)
PERT	passivated emitter, rear totally-diffused
PLD	pulsed laser deposition
PPy	polypyrrole
PTFE	polytetrafluoroethylene (Teflon)
PV	photovoltaic
PVD	physical vapour deposition
RF	radio frequency
SAED	selected-area electron diffraction
ScSZ	scandia-stabilised zirconia
SDC	samarium-doped ceria

SE	secondary electrons
SEM	scanning electron microscope
SHJ	silicon heterojunction
SOEC	solid oxide electrolysis cell
SOFC	solid oxide fuel cell
SWCNT	single-walled carbon nanotube
TCO	transparent conducting oxide
TEM	transmission electron microscope
TOPCon	tunnel oxide passivated contact
UNSW	University of New South Wales
UV	ultraviolet
wt %	percent by weight
XRD	X-ray diffraction
YSZ	yttria-stabilised zirconia